

JGR Atmospheres



RESEARCH ARTICLE

10.1029/2025JD045235

Key Points:

- We present 43 Terrestrial Gamma-ray Flashes (TGFs) with reliable locations detected by the ALOFT flight campaign
- We present the TGF photon brightness distribution and find that the distribution follows a power-law with a roll-off at lower brightness
- TGFs can be five orders of magnitude dimmer than previously thought and at least 97% of TGFs detected by ALOFT cannot be detected from space

Supporting Information:

Supporting Information may be found in the online version of this article.

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Citation:

Fuglestad, A. N., Marisaldi, M., Sarria, D., Mezentsev, A., Østgaard, N., Engeland, I. B., et al. (2026). The source brightness distribution of terrestrial gamma-ray flashes from the ALOFT flight campaign. *Journal of Geophysical Research: Atmospheres*, 131, e2025JD045235. <https://doi.org/10.1029/2025JD045235>

Received 22 AUG 2025

Accepted 21 FEB 2026

The Source Brightness Distribution of Terrestrial Gamma-Ray Flashes From the ALOFT Flight Campaign

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Abstract Terrestrial Gamma-ray Flashes (TGFs) are naturally occurring phenomena that consist of bursts of gamma radiation associated with thunderclouds. Measurements of TGFs have been predominantly based on satellite instruments due to the significantly larger observation area compared to ground measurements. Due to atmospheric attenuation of the gamma rays between the TGF source deep in the atmosphere, there is a minimum brightness threshold TGFs must reach to be detectable from space. How many TGFs occur below this threshold and the minimum possible source brightness of a TGF is still not known. In this paper we show 43 TGFs with reliable locations detected by the Airborne Lightning Observatory for Fly's Eye Geostationary Lightning Mapper (GLM) Simulator (FEGS) and TGFs (ALOFT) flight campaign, which recorded observations using the NASA ER-2 aircraft at an altitude of 20 km, targeting gamma-ray glowing thunderclouds and TGFs. Monte Carlo simulations were used to estimate the source photon brightness of the TGFs. Based on the results for 37 of the 43 TGFs, it was found that the TGF source brightness distribution follows a power-law with index $\alpha = -0.45$ with min max $\alpha = (-0.29, -0.79)$ which hardens below 10^{14} , >100 keV photons at source. Additionally, it was found that at least 97% of the ALOFT TGFs cannot be detected from space, and TGFs can have source brightnesses down to five orders of magnitude lower than what is detectable from space.

Plain Language Summary Terrestrial Gamma-ray Flashes (TGFs) are short naturally occurring bursts of gamma-rays associated with strong electric fields within thunderclouds. Based on satellite and aircraft measurements TGFs have been thought to be associated with only a small fraction of lightning flashes and strong electric fields. In this paper we present a distribution of the TGFs gamma-ray source brightness based on measurements by an aircraft campaign over the Gulf of Mexico. Our results show that TGFs can be about five orders of magnitude dimmer than previously thought and that at least 97% of TGFs cannot be measured by current satellite instruments, indicating that TGFs are a much more common phenomenon than previously thought.

1. Introduction

Since the discovery of Terrestrial Gamma-ray Flashes (TGFs) by the Compton Gamma Ray Observatory (Fishman et al., 1994), TGF research has been predominantly based on satellite measurements. However, because of the attenuation of gamma rays within the atmosphere and the high altitude of observing satellites, only TGFs with a source brightness above a certain minimum threshold can be detected. Dwyer and Smith (2005) found from Monte Carlo simulations that about $\sim 10^{17}$ runaway electrons with >1 MeV at a source altitude of 15 km were needed in order to match the measured average fluences by the Reuven Ramaty High Energy Solar Spectroscopic Imager (RHESSI) at an altitude of 600 km. The first estimates of TGF source brightness was performed by Cummer et al. (2014), who found two TGFs at ~ 12 km altitude with $\sim 10^{18}$ runaway electrons at source. Based on the skewness of the fluence distribution from RHESSI, Collier et al. (2011) suggested that TGFs might follow a power law distribution, significantly increasing in number with decreasing source brightness until some threshold value where the distribution would rapidly approach zero or plateau. Given such a distribution, if the threshold value is significantly below $\sim 10^{17}$ source photons, a large population of TGFs would exist that cannot be detected by current satellite instruments.

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Satellite measurements of the power law index reported in literature are similar. Østgaard et al. (2012) found from a joint fit of RHESSI and Fermi Gamma-ray Burst Monitor (GBM) data a power law with $\alpha = -2.3 \pm 0.2$ with a roll off of the power law at lower fluences down to $\alpha = -1.7$. Tierney et al. (2013) and Marisaldi et al. (2014) confirmed that the measured fluence distribution follows such a power law, Tierney et al. (2013) found from the Fermi GBM a power index of $\alpha = -2.20 \pm 0.13$ and Marisaldi et al. (2014) found from the Astro Rivelatore Gamma ad Immagini Leggero (AGILE) a best fit with index $\alpha = -2.4$. The fluence distribution at satellite altitude is different from the intrinsic source brightness distribution, as the first is a measure of total number of photons per unit area incident on the detector, while the second is the total number of photons emitted from the source. Based on a modeling study by Carlson et al. (2012), Østgaard et al. (2012) found the RHESSI fluence distribution to correspond to a source brightness distribution with $\alpha \simeq -2.0$ with a roll off to $\alpha \simeq -1.3$ at lower fluences. Østgaard et al. (2012) also stated that one could not rule out that there exists TGFs with $\sim 10^{12}$ electrons at source given the observations by RHESSI and Fermi and the non-detection by the Airborne Detector for Energetic Lightning Emissions (ADELE) (Smith et al., 2011a). Additionally Celestin et al. (2015) modeled high-energy photon emissions associated with lightning leaders and found that lightning leaders producing 300 MV potential drops lead to photon emissions consistent with TGFs, suggesting a close relation between X-ray bursts from lightning leaders and TGFs. These X-ray bursts can be about 10 orders of magnitude dimmer than the TGFs detected by satellite instruments, which suggests that TGFs detected by satellites may only represent a small fraction of all TGFs. A Monte Carlo simulation study connecting the fluence distribution observed from space with the source properties of TGFs is presented in Dwyer et al. (2017). Unfortunately, the observations available for this study cannot fully constrain TGF source properties.

The first observation of a TGF from an aircraft was done by ADELE, which detected one TGF over about 37 flight hours flying at an altitude of ~ 14 km around storms in the contiguous United States, mostly around Florida. ADELE was equipped with three kinds of scintillators 5×5 inch NaI, 5×5 inch plastic and 1×1 inch plastic, with an energy of ~ 50 keV to > 5 MeV (Smith et al., 2011b). Flying at this low altitude means that one can only circulate in the outskirts of the thunderstorms and not above tropical thundercloud tops. Based on the non-detections of the ADELE flight campaign, Smith et al. (2011a) estimated TGFs to be associated with only about 0.1%–1% of lightning flashes. Assuming that the fluence distribution of TGFs below satellite detection threshold also follows a power law, they estimate the power law index to be $\alpha > -0.8$. Further based on a stacking analysis of RHESSI data, Smith et al. (2016) concluded that there is no significant population of faint TGFs within four orders of magnitude of the detection threshold of RHESSI and found that the power law index must harden dramatically or that there must be a cutoff just below the sensitivity limit of current satellite instruments. Similarly, McTague et al. (2015) performed a lightning-based search for TGFs too dim to be detected in Fermi GBM and found no significant signal during the time windows in which TGFs were expected to occur and concluded that weaker TGFs are rarer than implied by simple extension of the power law to low fluence. This was contrasted by Østgaard et al. (2015) who used WWLLN to present a new population of weaker TGFs not identified by the RHESSI search algorithms, which suggest a larger population of weaker TGFs. A further study by Albrechtsen et al. (2019) found TGFs down to three counts in the detector, corresponding to 0.22 of the effective detection threshold ~ 13.625 counts of RHESSI (Smith et al., 2016). Additionally, they found no signs of a cutoff in the TGF distribution. More recent measurements have found further evidence for the existence of a population of low-fluence TGFs. Abbasi et al. (2018, 2024) and Kieu et al. (2024, 2025) found surface detections in the Telescope Array Surface Detector of downward propagating TGFs consistent with source brightnesses between $10^{11} - 10^{14}$ photons at source with energy > 100 keV. In 2023 the Airborne Lightning Observatory for Fly's Eye Geostationary Lightning Mapper (GLM) Simulator (FEGS) and TGFs (ALOFT) flight campaign (Lang et al., 2025; Marisaldi et al., 2024; Østgaard et al., 2024) resulted in the detection of more than 100 TGFs, from which (Bjørge-Engeland et al., 2024) reported six TGFs detected by ALOFT during an overpass of the Atmosphere-Space Interactions Monitor (ASIM). None of the TGFs were detected by ASIM which offers direct evidence of TGFs too weak to be detected by satellite instruments. Bjørge-Engeland et al. (2024) estimated the TGFs to have source brightnesses in the order of $10^{12} - 10^{16}$ photons with energy > 100 keV assuming a source at 15 km altitude.

In this paper, we present 43 TGFs from the ALOFT flight campaign that were geo-located with the Vaisala global lightning detection network (GLD360). Of the many instruments onboard the aircraft, here we use only the BGO gamma-ray detector and the FEGS optical instrument, described in Section 2. Using Monte Carlo simulations, we

estimate the source photon brightness (from here on referred to as source brightness) of these TGFs, build a source brightness distribution, and discuss it in the frame of previous satellite observations.

2. Instruments and Data

The ALOFT ER-2 aircraft was equipped with the UiB BGO instrument used for gamma-ray measurements in this study, consisting of three Bismuth-Germanium-Oxide (BGO) scintillator bars each one connected to a photo-multiplier tube. This is the same detector and readout system as used by Østgaard et al. (2019). Each bar has dimensions $150 \times 150 \times 32 \text{ mm}^3$, a 28-ns time resolution, and covers the energy range of 300 keV to 40 MeV. The data acquisition system is able to identify saturation and flag when counts start overlapping (Østgaard et al., 2019). The flag system divides counts into three different categories: normal, fast, and overflow. Normal counts have no pile-up, meaning that the readout returns to baseline before any subsequent counts. Fast counts occur on the tail of an earlier count before the readout has returned to baseline. As fast counts occur on the tail of a previous count, their energies are unknown. Overflow counts occur when the readout is saturated to the point where it is not possible to identify valleys between counts. For overflow counts, the system reports the duration until the readout drops below saturation. During the overflow count both the energy and amount of counts occurring are unknown.

Another set of gamma-ray detectors on board the ALOFT ER-2 aircraft were the Lutetium-Yttrium Orthosilicate (LYSO) detectors (Østgaard et al., 2019). There are three differently sized detectors. The large, medium, and small detector with dimensions: $50 \times 50 \times 50$, $10 \times 10 \times 10$, and $3 \times 3 \times 10 \text{ mm}^3$. The medium LYSO was used in this study in case the UiB BGO was saturated, to verify if saturated TGFs consisted of one or multiple pulses.

FEGS (Quick et al., 2020) is a multi-spectrum array of radiometers and electric field change meter. The array consists of twenty-five 777-nm radiometers arranged in a 5×5 grid. Each radiometer has a square field of view with a half angle of 9.46° and is positioned with an angle of 18° from the neighboring radiometers. The effective field of view of the whole array is approximately a square with a half angle of 45° pointing toward nadir. For a cloud top at 15 and 17 km this corresponds to the field of view covering areas of about $10 \times 10 \text{ km}^2$ and $6 \times 6 \text{ km}^2$, respectively. The 777-nm radiometers have a central wavelength of 780 nm and a full width at half maximum (FWHM) of 10 nm. These radiometers are used for observing the OI line emission at 777.4 nm associated with the hot lightning channel. In addition, for the 2023 flight campaign, FECS was equipped with five more radiometers all pointing toward nadir with central wavelengths of 340, 500, 870, 1,600, and 300–1,100 nm (Lang et al., 2025). The 340-nm radiometer were also used in this paper as they cover the N2 line emission at 337.1 nm which is part of the Second Positive band group and is associated with streamers. The 340-nm radiometer has a FWHM of 10 nm. All of the radiometers have a temporal resolution of 10 μs (Quick et al., 2020).

Vaisala global lightning detection network (GLD360) (Demetriades et al., 2010; Said & Murphy, 2016) is a network of radio detectors covering the very low frequency range (500 Hz–50 kHz). The network uses time of arrival (TOA) and magnetic direction finding to geo-locate lightning flashes. The network has a median accuracy of $\sim 2 \text{ km}$ and does not distinguish between whether a stroke was intra-cloud (IC) or cloud to ground lightning. The Gulf of Mexico where the ALOFT campaign took place, is a region where the network has a 1:1 total flash ratio when compared to the Optical Transient Detector and Lightning Imaging Sensor (Said & Murphy, 2016). An uncertainty range is provided for each detection in the form of an ellipse.

3. Method

3.1. TGF Identification and Geolocation

TGFs were identified by manual search in the BGO data as a short (10–100 μs) increase in counts above the background (0.0239 counts/10 μs or 2390 counts/s). The medium LYSO data was used as supplementary data in the search. From the manual search, 129 TGFs were found, 43 of which had an associated GLD360 detection. GLD360 detections were marked as associated if the detection occurred within 30 km of the aircraft and within $\pm 200 \mu\text{s}$. The 30 km distance was selected as a large bound to ensure no likely associated GLD360 detection were missed. FECS data was used as supplementary to the GLD360 detection and checked for any optical pulses in the 777 and 337 nm bandwidth, with onsets occurring within 3 ms of the TGF.

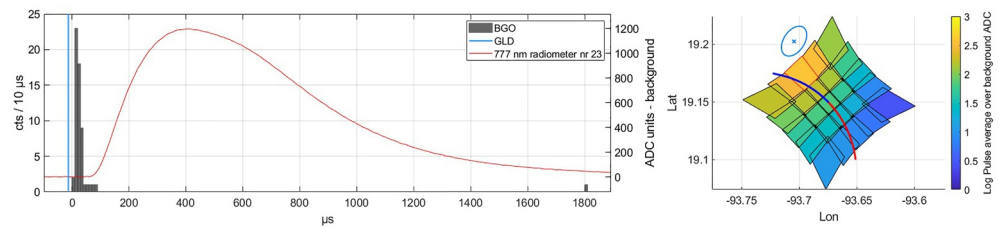


Figure 1. Terrestrial Gamma-ray Flashes (TGF) 24 07:07:17.304 UTC. Left panel shows a timeline of the TGF. The time starts at the onset of the TGF. BGO counts are indicated by the black histogram. The blue vertical line indicates the time of the GLD360 detection and the red line plot shows the ADC readout minus background of the 777-nm radiometer with the highest signal above background. The right panel shows a map during the TGF, the red line indicates the forward travel direction of the aircraft while the blue line indicates the flight path the aircraft traveled before the TGF. The blue ellipse shows the GLD360 detection associated to the TGF. The colored square shows the FoV of the 777-nm radiometer array projected onto a 15 km cloud top. The brightest radiometer is highlighted with a red border.

For TGFs with both GLD360 and FEGS data, the location of the GLD360 detection was compared with the direction of the optical signal indicated by the FEGS 777-nm array. Due to the positioning of the 777-nm photometers, it is possible to create a 5×5 pixel image of the optical emission occurring near the time of the TGF. The same approach to imaging was used in Bjørge-Engeland et al. (2024). The image is geolocalized using aircraft position and direction of motion assuming a cloud top altitude of 15 km, which is a reasonable assumption based on the median cloud top height measurements from the Cloud Radar System (Lang et al., 2025). In cases where the actual cloud top is higher, the image projection will be larger than the true footprint. The aircraft's 170 ms GPS data is linearly interpolated to obtain the location and direction of motion at the onset of each TGF. The brightness of each pixel in the image is determined by taking the common logarithm of the average value of the optical pulse over the background value. The time window of the optical pulse is manually determined for each pulse to ensure that the full pulse energy is included. The background value of each photometer is determined by taking the average value of 5 ms before the onset of the optical pulse, while making sure that there is no other optical activity occurring in this time window. The image is used to verify if the direction of the optical pulse corresponds to the location given by GLD360. Figure 1 shows an example of TGF, optical signal, and geolocation.

3.2. Monte Carlo Simulations

A library of Monte Carlo simulations run on a 1-km spaced grid of radial distances (0–15 km) and a production altitude of 15 km were used to quantify the source brightness of the TGFs. The Monte Carlo simulation can be broken down into four steps.

In the first step the simulation generates a TGF assuming a point source Relativistic Runaway Electron Avalanche (RREA) photon spectrum of $N = 10^8$ initial photons with energies in the range 100 keV to 40 MeV and a Gaussian angular distribution of sigma 30° pointing upwards with no tilt on the TGF emission direction. The equation used for the RREA photon spectrum is given by

$$f(E) = \frac{1}{E} \exp\left(\frac{-E}{7.3\text{MeV}}\right), \quad (1)$$

where E is the photon energy.

In the second step the photons are propagated from the reference altitude of 15 km to the aircraft altitude at 20 km.

In the third step the Monte Carlo simulations apply the response matrix of the BGO detector. The response matrix simulates the amount of photons and their energies that are deposited in the BGO based on a 3d model of the BGO and the surrounding aircraft, while also accounting for the energy detection threshold of the BGO. The results of which is a simulated recorded spectrum.

In the fourth step the Monte Carlo simulation compares the observed spectrum from the BGO data to the simulated recorded spectrum and scales the simulated spectrum by a factor M/N to best fit the observed spectrum,

where M is the resulting estimated source brightness. This scaling assumes that the $M - N$ additional photons will behave exactly as the initial $N = 10^8$ photons. The uncertainty of scaling the simulated spectrum to fit the observed spectrum is disregarded due to being insignificant compared to the uncertainty due to the radial location of the TGF. As the energy of the counts are necessary to create the observed energy spectrum, only the normal counts are used for the spectral fit. However, normal and fast counts are accounted for when scaling the source brightness. The estimated source brightness includes all photons with source energies above 100 keV.

A maximum likelihood analysis was performed. However, due to the low number of counts in the TGFs (the median count is 74 per TGF), the accepted intervals of the analysis were typically very large and therefore not useful in constraining the source location of the TGFs.

The maximal radial distance of 15 km is a computer power limitation to get a good signal-to-noise ratio.

The 30° Gaussian beaming angle is consistent with measurements of TGFs detected by satellite instruments and with analysis by other authors. Hazelton et al. (2009) found that a beaming angle of 45° half-angle cone best fits spectral analysis of RHESSI TGFs. Based on modeling of beaming types by Lindanger et al. (2021), this is about equivalent to a Gaussian angular distribution with sigma 35°. Additionally, Belz et al. (2020) measured a 35° half-angle cone for a downwards propagating TGF, which would be equivalent to a Gaussian angular distribution of sigma 25° (Lindanger et al., 2021). Further, it was found that a larger angular distribution, sigma $\geq 20^\circ$ was a more probable explanation of a non-detection by ASIM of a TGF observed by ALOFT during an ISS overpass (Björge-Engeland et al., 2024).

In this work we do not consider a possible tilt angle for the TGFs. This is because of the following reasons: First, based on the ALOFT measurements we have no way to establish the tilt of the observed TGFs. Second, as the RREA electrons are accelerated by the large scale electric fields within clouds, the tilt of the TGF is dependent on the tilt in the electric field within the cloud which we generally expect to be small with respect to the vertical direction. Lastly, accounting for the tilt in simulation requires applying assumptions on the TGF tilt distribution. As there is no current measure of the TGF tilt distribution any assumptions on the distribution would be arbitrary.

Dwyer and Smith (2005) found that for the average RHESSI TGF 2×10^{17} runaway electrons at a source altitude of 15 km were needed to be detectable by RHESSI. To compare, we use a reference altitude of 15 km for all our TGFs. The estimated source brightness is highly dependent on the horizontal distance from the TGF source to the aircraft, varying by orders of magnitude by changes in the horizontal distance of only a couple of km (see Figure 3 in Björge-Engeland et al. (2024)). Therefore, we select only the TGF with an associated GLD360 detection (detection within $<200 \mu\text{s}$, $<30 \text{ km}$), as this gives a sample of TGF with known radial locations. Due to the uncertainty of the GLD360 location for each TGF we get a span of possible source brightnesses for each TGF.

3.3. Building the Source Brightness Distribution

In order to build the source brightness distribution, the usual approach would be to make a histogram of TGFs based on the estimated source brightness from the Monte Carlo simulations. However, based on the results of the Monte Carlo simulations it was found that the source brightness of each TGF can span multiple orders of magnitude depending on the radial distance of the source. Therefore, it is impossible to bin these TGFs by source-brightness without TGFs occupying multiple bins. To account for this we try to determine the probability of a TGF occurring inside a source-brightness bin. A simple explanation of the method is described below and a thorough explanation can be found in Appendix A.

For each TGF the estimated source brightness from the Monte Carlo simulation is dependent only on the radial distance from the aircraft of the TGF source location. Hence the probability of a TGF occupying a source-brightness bin is equal to the probability that the true source position of the TGF is within the radial distance that corresponds to that source-brightness bin.

Here we apply an assumption on the probability distribution function (pdf) of the TGF source location:

1. The true source position of the TGF is within the GLD360 uncertainty ellipse.
2. Any location within the GLD360 uncertainty ellipse is equally probable to be the true source location of the TGF.

These assumptions are based on that GLD360 uses multi station TOA to geolocate lightning flashes (Said & Murphy, 2016). The geolocation is found by solving an optimization problem of a set of linear equations based on the TOA of the detected pulses in each station. The pulses detected by the stations have a certain instrumental error in the TOA which leads to an area of valid solutions to the optimization problem, in which all points inside the area are regarded as equally valid solutions. The solution area has a complex shape that depends on the particular observational geometry of the receivers contributed to the specific detection. This shape is approximated by an error ellipse, which is reported as a tolerance area of the provided solution. As the exact solution of the optimization problem is not known assuming a flat probability distribution inside the ellipse is a more conservative estimate than assuming a weighted distribution such as a bivariate normal distribution.

The probability of a TGF occupying a source-brightness bin is then found by integrating the pdf numerically over the radial distances from the aircraft corresponding to the source-brightness bin. The result of the method is that the probability of a TGF occupying a source-brightness bin is based on the ratio between the area of the GLD360 ellipse that source-brightness bin occupies relative to the total area of the GLD360 ellipse. An example of this is shown in Figure 2.

Applying the method for all TGFs in our data set (see Table 1), results in each source-brightness bin having a set of probabilities $p_1, p_2, \dots, p_n$ that correspond to the chance that each TGF is inside that source-brightness bin. A set of probabilities like this is a Poisson Binomial distribution, from which the expected value and confidence interval can be determined.

Lastly, we account for the detection threshold of the BGO detectors. For each source-brightness bin there is a certain radial distance in which TGFs can be distinguished from the background of the BGO detector (see Figure 3). This means that the maximum observation distance of the BGO detector is dependent on the intrinsic brightness of the TGFs, and that our distribution will be biased toward bright TGFs if we do not account for the radial distance to source from where the TGFs can be detected. This is similar to the Malmquist bias in astronomy where less luminous stars cannot be observed to the same distance as brighter stars. To account for this bias, we divide the expected value and confidence interval values of each source-brightness bin by the area TGFs can be detected within, for that source-brightness bin. The excluded area of each source-brightness bin is indicated in gray in Figure 3. The resulting source brightness distribution is shown in Figure 4.

4. Results and Discussion

Table 1 shows all TGFs that were found to have an associated GLD360 detection, occurring within $<200 \mu\text{s}$ of the onset of the TGF, and within $<30 \text{ km}$ of the aircraft. Of the 43 TGFs 6 have been excluded from the analysis, due to uncertain or no estimates of the source brightness.

Four of the excluded TGFs had associated GLD360 detections that were completely outside of the Monte Carlo simulation range of 15 km . These TGF are noted as “Out of range” under the estimated source brightness in Table 1. If the assumption of being upward pointing with no tilt is true, considering the decrease in observed fluence versus radial distance, these TGFs would be brighter than some of the brightest TGFs observed from space (Gjesteland et al., 2015; Smith et al., 2020) if their source were at the maximum distance allowed by the GLD location error. Given the rarity of these extremely bright TGFs, we deem this scenario unlikely. A more likely explanation for these cases is that the flash rate in the thundercloud was high enough that the association with GLD360 is an accidental coincidence, that the TGF had a significant tilt or a that the TGF had a lower source altitude and a tilt. Belz et al. (2020) found for a downwards propagating TGF a tilt of up to 45° from vertical. It is reasonable to assume that similar tilts are possible for upwards propagating TGFs. How these TGFs would affect the source brightness distribution would require determining for each TGF whether they are tilted or coincidentally associated with the GLD360. If the GLD360 association was coincidental, we would not have any constraint on the source location of the TGF and therefore cannot estimate how bright these TGF are. If the TGF are tilted we would have an extra free parameter to account for and would be at risk of overfitting these TGF in addition to requiring more computation power to estimate the source brightness beyond 15 km . Therefore, we cannot determine how these four TGF would affect the source brightness distribution if included in the analysis.

The last 2 of the excluded TGF were saturated TGFs. These are TGFs with at least 1 overflow count in the BGO detector, and are indicated with a * mark in Table 1. During a saturated count both the energy and amount of counts occurring in the BGO is unknown. Since the Monte Carlo simulation uses the normal and fast counts to

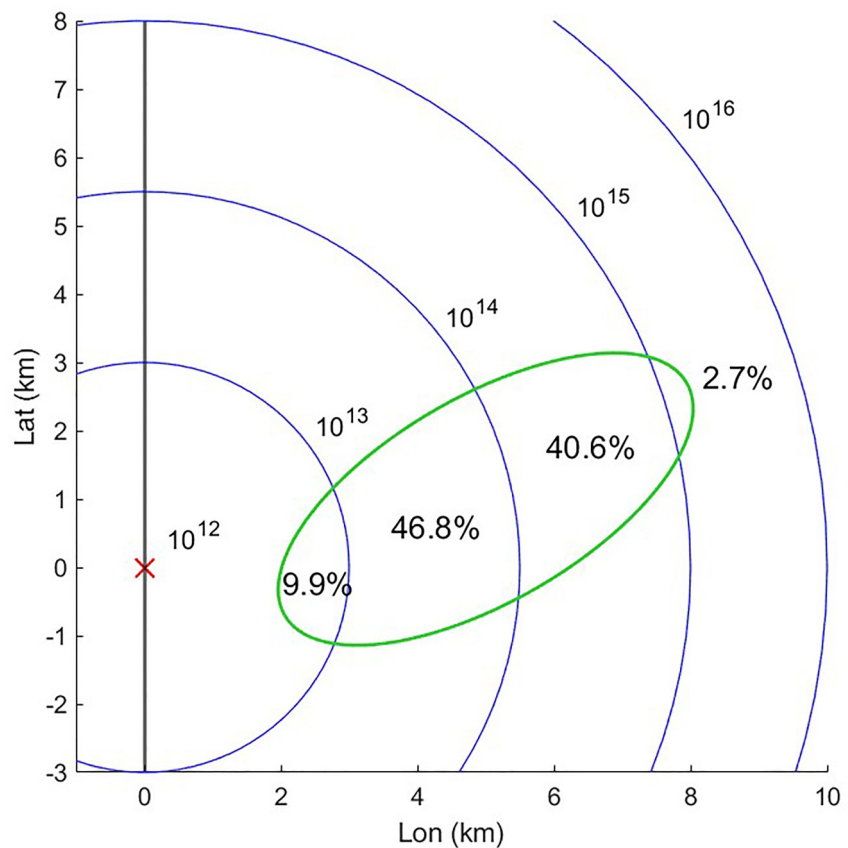


Figure 2. Illustrative figure of the method to determine the chance that a Terrestrial Gamma-ray Flashe (TGF) occurs within any source brightness bin. The Black line represents the flight path of the aircraft, while the red cross represents the position of the aircraft at the time of the TGF. The green ellipse shows the GLD360 uncertainty ellipse of the TGF. The blue circles show at what distance the estimated source brightness of the TGF would correspond to an order of magnitude. Since we apply the assumption that the true source position of the TGF is within the GLD360 ellipse and that any position inside the ellipse is equally probable, this means the chance of the TGF being in any source brightness bin is given by the ratio of the corresponding part of the ellipse to the total area of the ellipse. For example, in the figure, 46.8% of the ellipse is within the radial distance corresponding to the $10^{13} - 10^{14}$ bin.

estimate the source brightness we know that these TGF are underestimated. The medium LYSO was checked for these TGFs and confirmed that these TGFs consist of one pulse. We can get a rough estimate of how much these TGFs are underestimated by comparing the relative amount of counts in the LYSO detector to the BGO counts while scaling for the difference in effective area, which is about 225 times smaller for the LYSO than the BGO. Doing so we get a scaling factor of 137.2 and 68.8 for TGF 24 07:06:58.110 UTC and TGF 29 20:47:18.670 UTC, which corresponds to source brightness in the range $7.2 \times 10^{15} - 1.4 \times 10^{20}$ photons at source and $2.3 \times 10^{16} - 3.3 \times 10^{16}$ photons at source. If these events were included into the source brightness distribution they would increase the values for the upper source brightness bins from the $10^{13} - 10^{14}$ bin and up, if using the underestimated values and from the $10^{15} - 10^{16}$ bin and up, if using the LYSO adjusted estimates. We choose not to include these TGFs in the source brightness distribution as they account for only about 5% of the total TGFs if included and because the LYSO adjusted source brightness values are a rough estimate and should not be relied upon. However, these TGF show that we are slightly underestimating the upper bins of the distribution.

Out of the 37 remaining TGFs only one was found to have an estimated source brightness above the space detection threshold of 10^{17} source photons >100 KeV; for all possible radial distances.

Figure 3 shows the estimated source brightness versus radial distance for each TGF with an associated GLD360 geolocation. A notable feature is that the estimated TGF brightness can span several orders of magnitude depending on the true radial distance of the TGF. Additionally all TGFs appear to be in an almost logarithmically

Table 1
Terrestrial Gamma-Ray Flashes With Temporally Associated GLD360

Day/time (UTC)	Aircraft, lat/lon	GLD360, lat/lon	Δt (μ s)	Counts	Radial distance (km)	Est. source brightness, min/max
06 05:01:49.536	19.29, -94.36	19.29, -94.36	-20	83	0.00–2.10	8.24e11–5.12e12
06 05:01:56.650	19.29, -94.35	19.31, -94.42	-30	29	7.82–9.90	1.82e14–1.27e15
06 05:01:57.898	19.29, -94.34	19.31, -94.39	10	27	4.03–6.39	5.42e12–4.47e13
06 05:01:58.256	19.29, -94.34	19.32, -94.34	-20	28	1.39–5.04	8.22e11–1.11e13
06 05:02:01.448	19.29, -94.34	19.34, -94.31	20	125	5.24–7.81	7.84e13–9.26e14
06 05:07:38.047	19.29, -94.34	19.24, -94.41	-30	32	6.02–13.88	3.43e13–2.62e16
06 05:07:39.447	19.29, -94.34	19.27, -94.37	-20	33	0.40–6.78	4.05e11–7.14e13
06 05:07:56.844	19.27, -94.37	19.27, -94.38	10	195	0.08–2.87	2.11e12–1.92e13
06 05:08:26.306	19.23, -94.41	19.28, -94.39	-30	89	4.57–8.14	2.46e13–7.19e14
06 05:08:37.857	19.21, -94.42	19.30, -94.42	-50	43	7.97–11.29	2.30e14–4.34e15
06 06:52:00.307	19.40, -94.49	19.46, -94.43	20	57	8.14–11.15	4.88e14–7.03e15
08 05:45:48.534	13.20, -89.68	13.16, -89.71	-130	261	3.29–12.20	3.18e13–7.94e16
20 23:24:49.165	27.04, -82.05	27.00, -82.03	100	943	4.67–5.01	3.64e14–5.12e14
24 05:17:31.757	18.92, -94.42	18.95, -94.55	-60	180	13.40–16.25	1.13e17–3.01e17
24 05:17:48.503	18.91, -94.45	18.93, -94.42	20	67	2.08–5.83	3.02e12–5.59e13
24 05:17:55.498	18.90, -94.46	19.16, -94.47	80	55	22.85–35.03	Out of range
24 05:37:33.898	19.18, -94.69	19.19, -94.62	10	401	6.20–8.98	6.18e14–9.22e15
24 05:49:08.980	19.24, -94.33	19.40, -94.34	30	63	16.55–19.08	Out of range
24 06:08:10.151	19.33, -94.62	19.35, -94.56	-20	78	5.33–7.14	5.18e13–7.73e14
24 06:41:14.539	19.24, -94.01	19.28, -94.04	10	48	3.80–6.23	7.36e12–6.20e13
24 06:44:09.774	19.19, -93.73	19.11, -93.64	-30	134	10.98–14.91	1.10e16–1.55e17
24 06:45:28.422	19.10, -93.62	19.31, -93.66	90	101	18.08–29.94	Out of range
24 06:45:38.618	19.08, -93.60	19.10, -93.63	20	68	1.09–10.27	2.04e12–4.51e15
24 06:45:48.174	19.07, -93.59	19.06, -93.64	10	72	4.16–6.89	1.69e13–2.03e14
24 06:55:49.470	19.30, -94.08	19.31, -94.10	20	34	1.65–3.86	1.83e12–7.92e12
24 07:05:53.468	19.20, -93.82	19.19, -93.79	10	62	2.32–4.59	3.88e12–2.04e13
24 07:06:58.110*	19.17, -93.70	19.23, -93.73	80	741	2.59–14.59	5.28e13–9.87e17
24 07:07:17.304	19.15, -93.67	19.20, -93.70	-30	57	5.72–8.01	4.94e13–4.56e14
24 07:13:53.099	19.38, -94.13	19.35, -94.20	-50	65	6.94–10.07	1.74e14–3.35e15
24 07:24:48.973	19.22, -93.80	19.33, -93.81	20	128	11.00–13.59	1.35e16–8.34e16
24 07:27:10.606	19.13, -93.56	19.16, -93.49	20	59	7.05–9.68	1.86e14–2.28e15
24 07:27:31.426	19.12, -93.52	18.96, -93.51	-40	518	10.30–25.88	3.50e16–8.47e17
24 07:27:31.432	19.12, -93.52	19.06, -93.61	-20	43	7.15–16.07	1.48e14–7.37e16
24 07:27:36.225	19.12, -93.51	19.18, -93.50	10	184	1.47–15.56	5.96e12–2.80e17
24 07:32:45.975	19.19, -93.79	19.27, -93.76	20	46	7.26–10.63	1.37e14–3.08e15
24 07:40:50.906	19.27, -93.92	19.26, -93.89	20	147	2.64–5.01	1.13e13–6.90e13
24 07:41:10.403	19.25, -93.89	19.26, -93.90	10	91	0.11–2.41	7.84e11–5.92e12
28 17:14:27.984	15.45, -82.22	15.50, -82.19	50	74	4.08–7.11	1.59e13–2.51e14
28 17:14:27.984	15.45, -82.22	15.50, -82.19	-40	248	4.08–7.11	5.78e13–9.15e14
29 20:39:12.193	28.65, -80.91	28.66, -80.91	-20	92	1.11–1.51	2.80e12–3.53e12
29 20:40:11.584	28.76, -80.89	28.73, -80.81	-160	56	8.26–12.98	3.41e14–1.64e16
29 20:47:18.670*	28.64, -80.94	28.65, -80.89	-40	559	5.30–5.70	3.35e14–4.84e14
29 21:09:34.303	28.24, -80.92	28.42, -80.99	-40	239	21.49–22.98	Out of range

Note. Day/time indicates the day in the month of July 2023 and time at the onset of the TGF in UTC. Δt indicated the time difference from the onset of the TGF and the GLD360 detection in μ s. Radial Distance indicates the minimum and maximum radial distance from the aircraft to the GLD360 uncertainty ellipse. Estimated Source Brightness represents the minimum and maximum photons at source estimated by the Monte Carlo simulation based on the possible radial distance of the TGF, with a maximum estimation distance of 15 km from the aircraft. The TGFs marked with * indicate saturated TGF, these TGFs have at least 1 overflow count therefore the source brightness is underestimated.

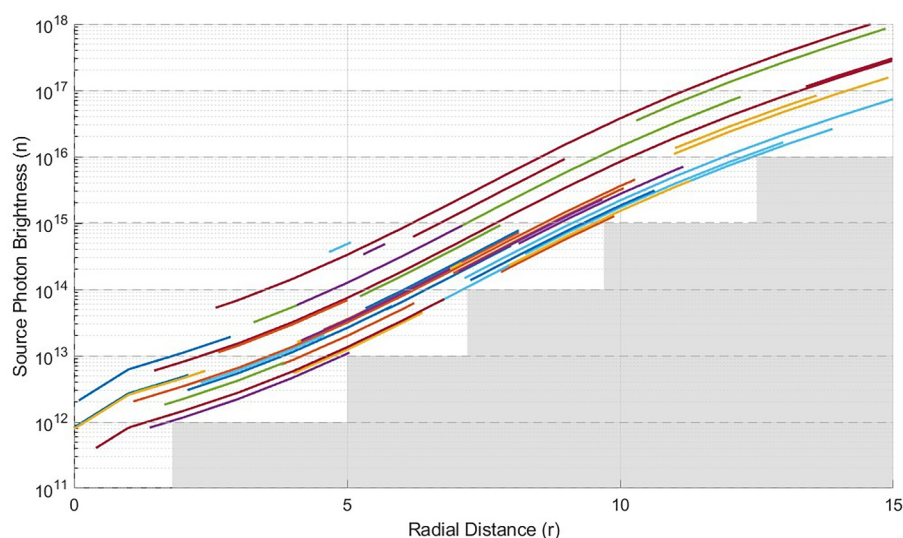


Figure 3. Source photon brightness versus radial distance of 37 of the Terrestrial Gamma-ray Flashes (TGFs) with associated GLD360. Each line in the plot represents one TGF with the radial distance corresponding to the possible source location of the TGF given the uncertainty in the GLD360 location. A reference altitude of 15 km is used for all TGFs in estimating the source brightness. The distribution stops at 15 km radial distance due to the radial limit of the Monte Carlo simulation. The gray area beneath the curve indicates the area at which the BGO experiences detection cutoff used in Figure 4.

linear band going from about 10^{12} photons for a source location at 0 km radial distance to about 10^{17} photons for a source location at 15 km radial distance. A likely explanation for this band is that TGFs below the band are too weak and too distant to be distinguished from the background in the BGO detectors. In contrast, TGFs above the band might be rarer due to their higher source brightness, and they could be capable of saturating the BGO detectors. Since saturated TGFs are excluded from the analysis it should be the case that the area these saturated TGFs occupy in the source-brightness/radial-distance space should be excluded from the area discussed in Section 3.3 in which TGFs of each source-brightness bin can be detected within. However, since we only have two saturated TGFs the exact boundary of this area is vague. Thus, we are likely underestimating the TGF source brightness density of the upper source brightness bins.

Figure 4 shows the estimated source brightness distribution, derived from the method described in Section 3.3. It is clear that TGFs with $10^{12} - 10^{13}$ photons at source dominate the TGF source brightness distribution. Considering the reference threshold for space detection of 10^{17} photons at 15 km source altitude, comparing the relative amount of TGFs above the threshold to the amount below we find from the expected values ~ 120 times more TGFs with source brightnesses below 10^{17} than above. If we assume a lower detection threshold of 10^{16} , which might be possible for ASIM, then we find ~ 33 times more TGFs with source brightnesses below 10^{16} than above. This suggests that the majority of TGFs cannot be detected by instruments onboard spacecraft, in agreement with the findings of Bjørge-Engeland et al. (2024). This means that assuming this set of TGFs is representative of tropical thundercloud TGFs at least 97% of the TGFs from tropical thunderclouds cannot be detected from space.

It is worth noting that due to the nature of the ALOFT flight campaign, the TGFs detected in this study are all upward-propagating TGFs. The source brightness distribution may not necessarily be the same for downward-propagating TGFs. Few measurements of downward-propagating TGFs have been made (Abbasi et al., 2018, 2024; Belz et al., 2020; Bowers et al., 2017; Enoto et al., 2017; Hare et al., 2016; Wada et al., 2019, 2025), and not enough to build a fluence or source-brightness distribution. However, the underlying RREA mechanism behaves the same regardless of the orientation of the electric field it occurs in. Because of this we would not expect any difference in the source brightness distribution of TGFs occurring in the upward and downward direction.

From Figure 4, it appears that the TGF source brightness distribution follows a power-law for source brightnesses in the range of $10^{14} - 10^{18}$, but the slope of the distribution decreases toward lower source brightnesses. A power-law fit of the $10^{14} - 10^{18}$ bins returned a best fit with power-law index $\alpha = -0.45$, with a minimum and

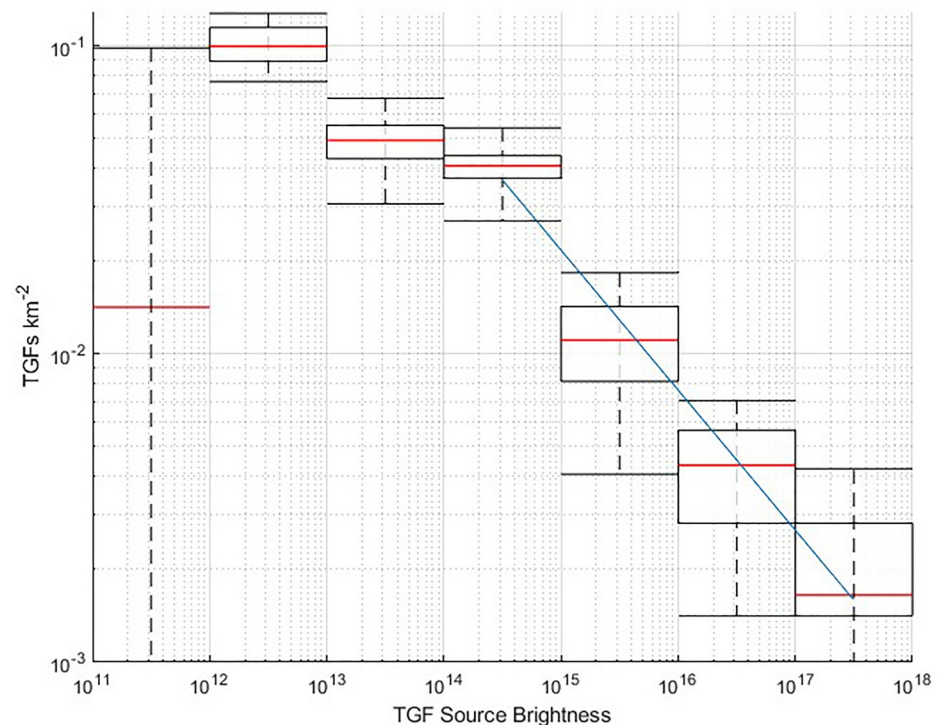


Figure 4. Log scale box plot of Terrestrial Gamma-ray Flashe (TGF) source brightness distribution. The values are drawn from a Poisson binomial distribution of 37 TGFs and represents the probability of having n TGFs within a source brightness bin divided by the corresponding detection threshold area of each source brightness bin. The upper boundary of the detection threshold area for each source brightness bin is indicated by the gray area in Figure 3. The red lines indicate the expected value for each source brightness bin, the box indicate the 50% confidence interval while the outer lines indicate the 95% confidence interval. The blue line indicates the best fit from a linear regression performed on the $10^{14} - 10^{18}$ bins, using the center of the bins as x -values. The unit of the y axis are TGF per km^2 over the entire flight duration. For the $10^{11} - 10^{12}$ bin the expected value contains the 50% confidence interval hence no box is drawn. For the $10^{11} - 10^{12}$ and $10^{17} - 10^{18}$ bin the lower 95% confidence interval value is at 0 and is therefore undefined on the Log scale.

maximum value of $\alpha = (-0.29, -0.79)$ corresponding to the least steep and steepest fit in the 95% confidence interval of the bins.

It is natural to compare our source brightness distribution with the corresponding results obtained from space observations. Both RHESSI, Fermi, and AGILE (Marisaldi et al., 2014; Tierney et al., 2013; Østgaard et al., 2012) report observed fluence distributions which can be fit with power laws with remarkably consistent index $\alpha = -2.2 \pm 0.2$. Note that these values are obtained considering several hundreds of TGFs detected from space, however all with source brightness likely above 10^{17} photons and resulting in about just one-order dynamic range in fluence. This is just the upper bin in the source brightness distribution from ALOFT data presented here, which extends over seven orders dynamic range down to 10^{11} photons at source, a range completely inaccessible to space observations. However, ALOFT results are based on 43 TGFs only and likely biased because of the design of the observational campaign, targeting highly convective storms in a narrow geographical region. In fact, we do not know if the distribution is representative of all meteorological conditions and geographical regions. Modeling studies (Carlson et al., 2012; Østgaard et al., 2012) have tried to connect the space observed fluence distribution with the TGF source brightness. Carlson et al. (2012) showed that a power law source brightness distribution with a small power law index, in absolute value, can be detected from space as a much steeper distribution (see Figure 5 in Carlson et al., 2012), and this behavior is not heavily affected by the source altitude distribution (see Figure 7 in Carlson et al., 2012). This reconciles the ALOFT source brightness power law index ($\alpha = -0.45$) with the space-observed fluence distributions ($\alpha = -2.2$). For large source brightness, a steepening or a cutoff in the source brightness distribution is also expected, at least for simple mathematical reasons, since the distribution would reach an unphysical number of TGFs if there is no threshold at which the distribution ceases to behave as a power law. Østgaard et al. (2012), using RHESSI data, suggests a roll-off in the slope of the observed fluence

distribution for low fluence values ($\alpha = -1.7$), from looking at Figure 5 in Carlson et al. (2012) one can see that a source index of ($\alpha = -0.45$) corresponds to observed fluence of about ($\alpha = -1.5$) for a sigma 30° Gaussian beam, further reconciling space observations with ALOFT results. Moreover, considering the single TGF detection reported by the ADELE flight campaign (Smith et al., 2011b), Østgaard et al. (2012) concludes that TGFs with intensity down to 10^{12} electrons at source cannot be excluded. Considering a ~ 0.1 factor between relativistic electrons and Bremsstrahlung photons (Skeltved et al., 2014), this value is consistent with the lowest source photon brightness reported by ALOFT in this work. However, we cannot exclude a selection bias preventing us from identifying part of the weak TGFs originating at larger radial distances. In this case, the distribution would be harder and the fraction of TGFs not detectable from space would be even larger.

5. Conclusion

From manual search of the ALOFT flight campaign data, 129 TGFs were found. We present a sample of 43 TGFs with reliable location based on both correlated optical pulses and lightning detection networks. With the estimate of the source location and gamma-ray observations from the ALOFT, Monte Carlo simulations were used to get an estimate of the source photon brightness of the TGFs, assuming RREA spectrum with a sigma 30° Gaussian emission angle with no tilt. The main conclusions are:

- TGFs can be much dimmer than the TGFs that so far have been detected by satellite instruments. TGFs can have a source brightness at least as low as 10^{12} to 10^{13} photons at 15 km source altitude with energies above 100 keV.
- TGF source brightness distribution appears to follow a power-law with index $\alpha = -0.45$ with min max $\alpha = (-0.29, -0.79)$ which hardens for source fluences below 10^{14} photons at source.
- Assuming the set of TGFs with reliable location are representative of tropical thundercloud TGFs, the population is at least 33–120 times larger than that detectable from space, meaning that at least 97% of the tropical thundercloud TGFs cannot be detected from space.

More work can be done to improve the accuracy of the source brightness distribution, measurements of the source altitude in addition to radial distance from the gamma ray detector would provide a much more accurate estimate of the source brightness of the TGFs. Additionally a similar source brightness distribution can be built for downwards propagating TGFs to investigate if there are any intrinsic differences between upward and downward propagating TGFs. Lastly a further investigation into the difference in characteristics between the TGFs that were associated with GLD360 versus those that were not, could determine if the source brightness distribution presented in this paper may be biased toward certain types of TGFs.

Appendix A: Statistical Approach of Binning Radial Distribution

As discussed in Section 3.3, there are two factors that complicate building the source brightness distribution.

1. Each Terrestrial Gamma-ray Flashes (TGF) can span multiple orders of magnitude in source brightness.
2. The Bismuth-Germanium-Oxide (BGO) has instrumental cutoff, which is dependent on the source brightness of the TGF.

We will address the instrumental cutoff at the end. Due to the large span of possible source brightnesses of each TGF it is impossible to bin the TGF in the source-brightness space without some TGFs occupying multiple bins. To account for this, we use a statistical approach to attempt to determine the probability of the TGF occurring in any bin. Because of the assumptions applied to the Monte Carlo simulations (RREA spectrum, sigma 30° Gaussian angular distribution, upwards directed with no tilt and 15 km source altitude) the only variable that effects the estimated source brightness of the TGFs is the radial distance from the aircraft. The possible span in radial distance stems from the uncertainty of the true source location of the TGF. We estimate the source location based on the GLD360 data, which provides an uncertainty ellipse for each detection. Here we apply an assumption on how the probability of the source location of the TGF is distributed over the GLD360 uncertainty ellipse, by defining a probability distribution function (pdf) for the source location of the TGF. In this case we assume:

1. The source position of the TGF is within the GLD360 uncertainty ellipse.
2. Each location within the GLD360 uncertainty ellipse is equally probable to be the true source location of the TGF.

From the assumption on the pdf, we can then determine a radial probability distribution, that is the pdf expressed as a function of radial distance from the aircraft. Then based on the radial probability distribution, we can calculate the probability of the TGF being inside any source-brightness bin. Repeating this process for each TGF, results in each bin having a Poisson binomial distribution from which the estimated amount of TGFs and the confidence interval can be calculated.

A1. Probability Distribution

Based on our two assumptions the pdf can be expressed as:

$$f(x,y) = \begin{cases} c, (x,y) \in A \\ 0, (x,y) \notin A \end{cases} \quad (\text{A1})$$

Where f is the pdf, A is the area of the GLD360 uncertainty ellipse, (x,y) corresponds to longitude and latitude and, c is some constant value.

That is to say, the probability is some constant value inside the area of the GLD360 uncertainty ellipse and zero outside. As the total probability must be equal to 1, the integral over the pdf for all values must also be equal to 1.

$$\int_{\mathbb{R}^2} f = 1 \quad (\text{A2})$$

Since the pdf is equal to zero outside of A , this is equivalent to:

$$\int_A f = 1 \quad (\text{A3})$$

The radial probability distribution can then be found by expressing the pdf as a function of the radial distance from the aircraft.

The aircraft at the time of the TGF is at some location (x_0, y_0) , if we construct a circle with the aircraft location as the center with a radius r from the aircraft, we can then sample the value of the pdf for any angle θ along the circle. Integrating the pdf at the radius r over all possible angles of θ then gives the probability of the TGF being at a distance r from the aircraft, which corresponds to the radial probability distribution f_r at the radial distance r . Since the pdf is constant inside of A and zero outside, the integral is then equal to the length of the cross section between the circle and the ellipse multiplied by the constant c ,

$$\int_0^{2\pi} f = f_r = l(r)c \quad (\text{A4})$$

where $l(r)$ is the length of the cross section between the circle and the ellipse at the circle radius r .

$l(r)$ can then be calculated based on the geometry of the ellipse, and the radial probability distribution is then found by taking advantage of the fact that the integral over the radial probability distribution function over all radial distances r must be equal to 1:

$$\int_0^r f_r = \int_0^r l(r)c = 1 \quad (\text{A5})$$

$$c = 1 / \int_0^r l(r) \quad (\text{A6})$$

The radial probability distribution is given by:

$$f_r = l(r)c = l(r) / \int_0^r l(r) dr \quad (\text{A7})$$

From Equation A7, we then see the radial probability distribution can then be found simply by finding the cross section between the GLD360 ellipse and a circle drawn from the location of the aircraft for all possible radii r of the circle, where the circle and ellipse intersect.

Since the estimated source brightness of the TGFs are only dependent on radial distance, we can determine the corresponding radial-distance range for each source-brightness bin. Then the probability of the TGF occurring in any source-brightness bin is found by integrating the radial probability distribution over the corresponding radial-distance range of that source-brightness bin. Repeating this for all TGFs results in each source-brightness bin having a set of probabilities $p_1, p_2 \dots p_n$ that corresponds to the chance of each TGF being inside of that bin.

A2. Poisson Binomial

Each bin has a set of probabilities $p_1, p_2 \dots p_n$ which exist in the number space $p \in (0, 1)$. This distribution is a Poisson binomial distribution, which is similar to a binomial distribution except that every binomial trial has a different probability of a success.

The expected value of a Poisson binomial distribution is given by

$$E = \sum_{i=1}^n p_i \quad (\text{A8})$$

Since the Poisson binomial distribution is skewed, the confidence interval has to be found from the probability mass function. The probability mass function returns the chance of having k successful trials out of n total possible successes. The probability mass function can be calculated by using an iterative approach (Biscarri et al., 2018).

For the first iteration:

$$P_1(K=0) = q_1 \quad (\text{A9})$$

$$P_1(K=1) = p_1 \quad (\text{A10})$$

For subsequent iterations:

$$P_i(K=0) = P_{i-1}(K=0)q_i \quad (\text{A11})$$

$$P_i(K=k \neq 0, i) = P_{i-1}(K=k-1)p_i + P_{i-1}(K=k)q_i \quad (\text{A12})$$

$$P_i(K=i) = P_{i-1}(K=i-1)p_i \quad (\text{A13})$$

where P_i is the probability mass function for the set of probabilities $\{p_1, p_2, \dots p_i\}$ where $i \in (1, n)$, K is the number of successful trials, and $q_i = 1 - p_i$.

Calculating the probability mass function for each TGF that can be inside the bin from $p_1, p_2 \dots p_n$, P_n then returns the probability of having any amount of TGFs that are possible to have inside the bin. The confidence interval of 50% and 95% can be found to the nearest number of TGFs.

Lastly we account for the instrumental cutoff of the BGO. Each source-brightness bin has a different effective distance, in which TGFs in that source-brightness range can be distinguished from the background in the BGO, as seen in Figure 3. This means that simply comparing the expected amount of TGFs in each source-brightness bin would not account for the fact that TGFs with a higher source brightness can be observed from a much larger area. Therefore, we rather compare expected amounts of TGF per km^2 . To do so we simply divide our expected value and the value corresponding to the confidence intervals by the area TGFs of that source-brightness bin can be detected in. These areas are indicated in Figure 3 as the areas with white background.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The data used in this study are available on Zenodo via: <https://doi.org/10.5281/zenodo.16919087> (Fuglestad, 2025). The Monte Carlo simulation used in this study is available on Zenodo via: <https://doi.org/10.5281/zenodo.15349428> (Sarria, 2025b). The response matrix for the UiB-BGO used for the Monte Carlo simulations in this study is available on Zenodo via: <https://doi.org/10.5281/zenodo.15350194> (Sarria, 2025a).

Acknowledgments

This work made use of data from UIB-BGO, FECS, and the Global Lightning Detection Network. This project has received funding from the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (Grant 101201768). The ALOFT campaign and the UIB-BGO instrument were supported by the ERC under the European Union's Seventh Framework Programme (FP7/2007-2013, Grant Agreement No. 320839) and the Research Council of Norway (Contract Nos. 223252/F50 (CoE) and 325582). The simulations were performed on resources provided by UNINETT sigma2—the National Infrastructure for High Performance Computing and Data Storage in Norway, under project no. NN9526K. Work on ALOFT at the US Naval Research Laboratory is supported by the Office of Naval Research 6.1 funds. In addition, D. Shy is supported by the US Naval Research Laboratory's Jerome and Isabella Karle Fellowship. The FECS and EFCM team acknowledge the work of S. Podgorny, D. Corredor and M. Stewart. M.F. was sponsored by the Royal Society, UK (Grant NMGR/1/180252) and the Natural Environment Research Council, UK (Grant NE/L012669/1 and NE/H024921/1). Significant financial and logistical support for ALOFT was provided by the NASA Earth Science Division. We thank the governments of Mexico, the Bahamas, Colombia, Belize, Guatemala, Honduras, Nicaragua, Panama, the Dominican Republic, Costa Rica, El Salvador, Haiti, Turks & Caicos, Jamaica and the Cayman Islands for approving ER-2 overflights in support of ALOFT. We thank the ER-2 Project Team at NASA Armstrong Flight Research Center and the MacDill Air Force Base for acting as hosts.

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